

HERMES — 75-Second Prototype Demo Script

Flow: Operator → Engineer → Mission Sandbox → Operator → Maintenance

Time	View	What to Show	Purpose
0–15 sec	Operator	Live dashboard, telemetry, anomaly alert	Detect the anomaly
15–32 sec	Engineer	Fault diagnosis, parameters, root cause, RUL	Explain the anomaly
32–48 sec	Engineer	Mission Sandbox, current vs. what-if scenario	Simulate a safer mission profile
48–60 sec	Operator	Recommendation, Accept, updated RUL/risk	Make the decision
60–75 sec	Maintenance	Degradation trend, RUL, maintenance action	Close the loop

0–15 sec — OPERATOR VIEW

Show: Start on the Operator dashboard. Show the live telemetry, mission status, engine health/RUL, and the anomaly alert.

Speak: “We start with the operator. Hermes continuously monitors live engine telemetry against its physics-based digital twin. Here, it detects an abnormal deviation before conventional thresholds are breached.”

Action: Click the anomaly.

Continue: “The operator doesn’t just get an alarm — Hermes sends the issue for deeper diagnosis.”

15–32 sec — ENGINEER VIEW

Show: Switch to Engineer View. Show the anomaly type, relevant sensor/parameter graphs, fault classification, root-cause information, and current RUL.

Speak: “The engineer view explains what the anomaly actually means. Hermes separates sensor deviation from genuine engine degradation and identifies the likely fault.”

Action: Point at the diagnosis.

Continue: “Here, the model identifies the degradation pattern and estimates the engine’s remaining useful life.” Then immediately click Mission Sandbox.

32–48 sec — MISSION SANDBOX

Show: Show the current mission versus an alternative scenario. Change one mission parameter/profile and display the simulated risk/RUL comparison.

Speak: “Now Hermes goes beyond diagnosis. The engineer can test a ‘what-if’ mission scenario before the aircraft flies it.”

Action: Change the mission profile.

Continue: “Hermes fast-forwards the mission using the current engine health and predicts the resulting risk and RUL. This profile reduces the predicted engine stress, so Hermes recommends it as the safer alternative.”

48–60 sec — BACK TO OPERATOR

Show: Return to Operator View. Show the sandbox recommendation, before/after comparison, and Accept button. Then accept the recommendation and make the updated RUL/risk visible.

Speak: “The decision remains with the operator. They can review the recommendation and accept the alternative mission profile.”

Action: Click Accept and pause briefly for the values to update.

Continue: “Once accepted, Hermes updates the predicted mission outcome and RUL based on the selected scenario.”

60–75 sec — MAINTENANCE VIEW

Show: Switch to Maintenance View. Show degradation trend, RUL, fault history, and maintenance recommendation/priority.

Speak: “Finally, the same intelligence reaches maintenance. Instead of waiting for a calendar-based overhaul or a critical failure, technicians see the degradation trend and RUL-driven maintenance requirement.”

Action: End on the maintenance dashboard.

Continue: “So Hermes closes the loop: detect the anomaly, explain the fault, simulate the mission, make the decision, and plan maintenance — all from one digital twin.”

Quick Memorization Version

1. "We start with the operator. Hermes continuously compares live telemetry with its physics-based digital twin and detects this anomaly before a critical threshold is reached."
2. "The engineer view explains the deviation, identifies the likely fault, and estimates remaining useful life."
3. "Instead of stopping at diagnosis, Hermes opens the Mission Sandbox. We can test a what-if mission using the engine's current health and see how the predicted risk and RUL change."
4. "The operator reviews the recommendation and accepts the safer profile. Hermes immediately updates the mission prediction."
5. "Finally, maintenance receives the updated degradation trend and RUL, allowing maintenance to be planned before failure."
6. "Detect. Diagnose. Simulate. Decide. Maintain. That's Hermes."

Presentation Rule

Do not explain the algorithms during the 75-second demo. Keep the story focused on: **Something is wrong** → **Hermes understands it** → **Hermes tests the future** → **the operator decides** → **maintenance acts**.

Three screens: Operator = "Something is wrong." Engineer = "I know what it is and what could happen."
Maintenance = "I know what needs attention."